CryptoViz — AES Mode Visualizer

Detailed Project Report

Submitted By

• 23F-0557

• 23F-0720

• 23F-0624

• 23F-0569

Introduction

CryptoViz — AES Mode Visualizer is an interactive educational platform developed to help students understand AES encryption modes visually and practically.
Cryptography is one of the most important fields in cybersecurity because modern communication systems depend heavily on encryption to secure sensitive data.
Despite its importance, many students find cryptography difficult because most encryption processes are hidden and theoretical.

Our project solves this issue by creating a browser-based visualization system that demonstrates how AES encryption modes behave using real image encryption.
Instead of only reading theory, users can upload images and directly observe how encryption transforms them.

The project compares ECB, CBC, and CTR modes side-by-side and explains their differences through visual analysis, histograms, randomness testing, avalanche effect demonstrations, and decryption verification.

Problem Statement

Traditional methods of teaching cryptography rely mostly on lectures and textbook explanations.
Students often memorize concepts such as ECB weakness, CBC chaining, or CTR counters without truly understanding how these modes behave in practice.

This project was developed to provide a visual and interactive learning environment where users can observe encryption behavior in real time.
The aim is to simplify difficult cryptographic concepts using visualization and practical demonstrations.

Objectives

The major objectives of this project are:

1. To visually demonstrate AES encryption modes.
2. To compare ECB, CBC, and CTR using real image encryption.
3. To explain why ECB mode is insecure.
4. To demonstrate the importance of randomness in encryption.
5. To implement histogram analysis for encrypted images.
6. To implement avalanche effect demonstrations.
7. To verify encryption and decryption accuracy.
8. To create a fully frontend educational platform.
9. To help students understand modern cryptographic systems.
10. To provide an interactive cybersecurity learning experience.

Scope of the Project

The scope of CryptoViz focuses mainly on educational cryptography demonstrations.
The project allows users to upload PNG or JPG images, generate AES keys through SHA-256 password hashing, and compare multiple encryption modes visually.

The system focuses on:
• Encryption visualization
• Block analysis
• Histogram analysis
• Randomness analysis
• Avalanche effect testing
• Decryption verification

The project is not designed for production security systems.
Instead, it serves as a learning and research platform for students and cybersecurity enthusiasts.

Technologies Used

The project uses modern web technologies to create a responsive and interactive platform.

Frontend Technologies:
• HTML5
• CSS3
• JavaScript

Libraries and APIs:
• Chart.js for histogram visualization
• Web Crypto API for encryption and decryption
• Canvas API for image processing

Development Environment:
• VS Code
• Browser Developer Tools
• Local HTTP Server

These technologies allowed us to build a lightweight, fast, and interactive educational application.

AES Encryption Modes

AES (Advanced Encryption Standard) is one of the most widely used symmetric encryption algorithms in the world.
However, AES behaves differently depending on the encryption mode used.

ECB (Electronic Codebook):
ECB encrypts each block independently.
Identical plaintext blocks produce identical ciphertext blocks, which leaks patterns and makes ECB insecure for most applications.

CBC (Cipher Block Chaining):
CBC improves security by XOR-ing each plaintext block with the previous ciphertext block before encryption.
This introduces randomness and removes visible patterns.

CTR (Counter Mode):
CTR converts AES into a stream cipher using counters and nonces.
It is fast, parallelizable, and widely used in modern secure systems.

Our project visually demonstrates these differences through encrypted image comparisons.

System Features

CryptoViz contains several advanced educational features:

1. Real-Time Image Encryption
Users can upload an image and instantly view encrypted outputs for ECB, CBC, and CTR modes.

2. Block Pattern Analysis
The system highlights identical blocks using matching colors to demonstrate pattern leakage.

3. Histogram Analysis
Pixel frequency histograms are generated for each encryption mode to analyze randomness visually.

4. Avalanche Effect
Users can modify a single pixel and observe how encryption changes dramatically.

5. AI Randomness Analyzer
The system calculates entropy, chi-square values, and byte correlation to evaluate encryption quality.

6. Decryption Verification
Encrypted images are decrypted and compared with the original to verify correctness.

Working Methodology

The working process of the project follows these steps:

1. User uploads an image.
2. The image is rendered onto a canvas.
3. User enters a password.
4. SHA-256 converts the password into a 256-bit AES key.
5. The image data is encrypted using ECB, CBC, and CTR modes.
6. Encrypted outputs are displayed visually.
7. Statistical analysis is performed.
8. Histograms and block maps are generated.
9. Decryption verifies data integrity.

All operations are performed directly inside the browser using the Web Crypto API.

Challenges Faced

The team faced several technical and design challenges during development.

1. Browser Cryptography
Implementing encryption correctly using the Web Crypto API required careful handling of buffers and keys.

2. Image Processing
Managing image pixel data efficiently was challenging because large images consume significant memory.

3. Visualization
Creating visually understandable comparisons between AES modes required multiple design improvements.

4. Statistical Analysis
Implementing entropy and randomness analysis involved understanding cryptographic statistics deeply.

5. Responsive Design
Ensuring the interface worked smoothly on different screen sizes required additional CSS optimization.

Learning Outcomes

This project provided valuable learning experiences in multiple domains.

Cryptography:
• AES encryption
• Encryption modes
• Randomness analysis
• Avalanche effect

Frontend Development:
• Advanced JavaScript
• Canvas rendering
• Interactive UI design
• Responsive layouts

Cybersecurity:
• Secure encryption concepts
• Password hashing
• Statistical security analysis

Teamwork:

• Collaborative development
• Task management
• Debugging and testing

Future Improvements

Future versions of CryptoViz may include:

• Additional AES modes such as GCM and OFB
• File encryption support
• Video encryption visualization
• Real-time encryption speed benchmarking
• Machine learning based randomness analysis
• Exportable encryption reports
• User authentication system
• Multi-language support

These improvements would make the project even more powerful as an educational cybersecurity platform.

Conclusion

CryptoViz successfully demonstrates how AES encryption modes behave visually and interactively. The project transforms difficult cryptographic concepts into understandable visual experiences.

By combining encryption, image processing, statistical analysis, and interactive design, the platform provides a strong educational environment for students learning cybersecurity and cryptography.

The project also improved our technical skills in frontend development, browser cryptography, teamwork, and visualization systems.

Overall, CryptoViz achieved its goal of making encryption easier to understand through practical demonstrations.